

Practical task 2

1. Given the values of **P(B)** and **P(A and B)** in the following scenarios. You are welcome to calculate **P(A|B)** if you choose to do so:

- You work for a risk analysis insurer. You have read that this year, out of all drivers on the road, 5% have had accidents under the age of 25. You have also read that 10% of all drivers are under the age of 25. A new client approaches you and states that their age is 22. You want to calculate the chance that this driver has had an accident this year based on their age.

$P(A \text{ and } B) = 5\%$ of people under 25 AND had an accident

$P(B) = 10\%$ of all drivers are under 25

So the chance that they have had an accident is $0.05 / 0.10 = 0.5$ (So 50% probability)

- Your friend told you that they would buy you lunch if you can flip a coin and have it land on heads twice. You flip it the first time, and it lands on heads. What are your chances now of it landing on heads again?

$P(B) = 50\%$ chance that the first flip landed on heads

$P(A \text{ and } B) = 25\%$ chance of the second landing on heads

So the chances of landing on heads twice is $0.25 / 0.50 = 0.50$ (So 50% probability)

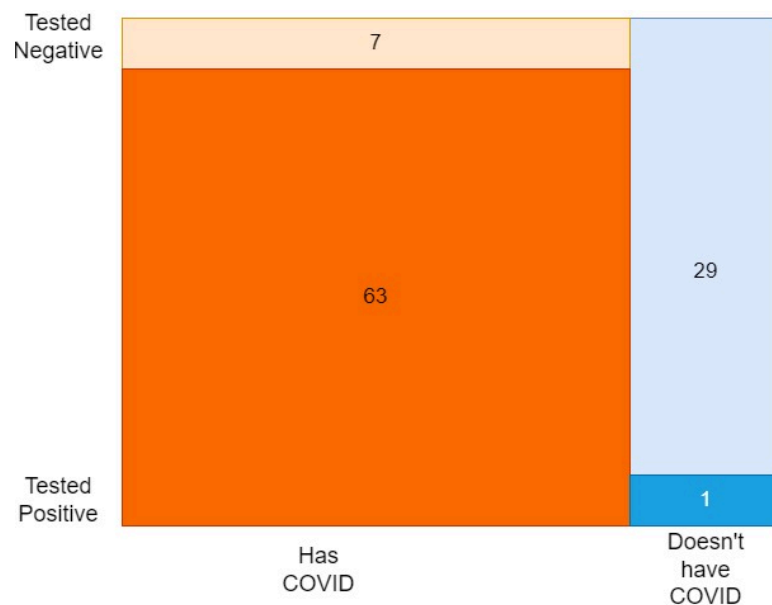
- You were always told that knowing Maths helps you to achieve 80% in Computer Science. You read some statistics showing that 30% of all Computer Science graduates took Maths and achieved 80%. Overall, 60% of all Computer Science graduates took Maths. Considering you took Maths, what are your chances of achieving 80%?

$P(A \text{ and } B) = 30\%$ took maths AND achieved 80%

$P(B) = 60\%$ took maths to begin with

So the chances of me achieving 80% is $0.30 / 0.60 = 0.5$ (So 50% probability)

2. Consider a mock study about COVID diagnosis with a total of 100 participants. A visual representation of participants and their COVID and diagnostic test result status is given below.



The two orange areas show the total number of people who have COVID, and the two blue areas show the total number of people who do not have COVID. The two darkly-coloured areas at the bottom show the people who tested positive for COVID. The two lightly-coloured areas show the people who tested negative for COVID.

- Using this diagram, state the following:
 - H: Someone has COVID (70 people)
 - E: Tested positive (64 people)
- Then, give the values for the following:
 - $P(H)$ - 0.70
 - $P(E|H)$ - 0.63 / 0.70
 - $P(E)$ - 0.64
 - $P(H|E)$ - 0.63 / 0.64